

In many cases, there could be more than one route from the router to the destination IP address range. To avoid loops and to find the most effective path to the destination, various routing protocols such as RIP, OSPF, EIGRP (Cisco Proprietary) and BGP are used.

In addition to routing the packets, a router may offer other services such as a DHCP server, a time server, DNS server, load balancing between two Internet connections, etc.

Default gateway: Logically, a default gateway is an IP address and subnet mask. It will always be a router, and it must communicate with at least two IP ranges.

Vyatta

The open source community offers a number of pleasant surprises. Ever imagined a router distro that can convert any computer to a router? In fact, Vyatta, an open source router distro, has been available since 2006! Though acquired by Brocade Communications in 2012, the Vyatta community edition is still available as an open source distro, though the latest commercial releases are no more open source.

As mentioned on vyatta.org, the free community Vyatta Core software (VC) is an award-winning open source network operating system providing advanced IPv4 and IPv6 routing, stateful firewalling, IPSec and SSL OpenVPN, and more. When you add Vyatta to a standard x86 hardware system, you can create an enterprise grade network appliance that easily scales from DSL to 10 GBps.

Vyatta deployment is possible on physical computer hardware, as well as virtualisation platforms such as VMWare, XenServer, Red Hat KVM and even Microsoft Hyper-V.

Let us try and understand how to install and configure Vyatta for a simple, real-life scenario. On the Internet side, the ISP has provided a static IP address. And the internal devices, connected to an ethernet switch use a private IP address range ... (Refer to Figure 1: network diagram) which can access the Internet using a Vyatta router. To install Vyatta in this scenario, a computer system with a minimum of two Ethernet cards, 512 MB RAM and 2 GB storage is required.

Installation

Get the latest ISO from vyatta.org, burn a CD and boot the system using this CD.

You can press **F1** on the first screen for installation help, or

Standard off-the-shelf router boxes are sufficient for slow speed Internet connections, typically up to 2 – 4 MBPS, but often, their performance drops if the available Internet bandwidth is more. Here, Vyatta comes handy. Additionally, it also provides VPN functionality, an SPI firewall and WAN load balancing.

It is also recommended that Vyatta core (Community) should be used only for testing and development environments, and users need to upgrade to the Vyatta subscription edition for production environments.

press **Enter** to continue. The Vyatta login prompt greets you here. The default login user is `vyatta` and the password is also `vyatta`.

The CD boots Vyatta in Live mode—to run it on a trial basis, without installing it on the hard disk. Here, you can configure the router to test its functionality.

System installation on the hard disk

To install the Vyatta image on the hard disk, enter the following command:

```
install system
```

This will start the installation of `vyatta` on the hard disk of the system. Next, select *auto partitioning* when prompted for selecting partitioning.

Now enter the name of the installation drive (`sda` is used in our installation). Next you will be prompted with a warning stating that *This will destroy all data on /dev/sda*. Be sure that you are okay with this before confirming as this process is irreversible and you may lose data if there is any on the harddisk. The details can be seen in figure 2.

Secure Vyatta with a strong password at the next prompt.

Figure 2: Installation Screen

As soon as the basic Vyatta installation is complete, remove the CD and reboot your system using the hard disk as a boot device. The computer will boot into the new world of Vyatta.

Configuration

To view the existing configuration, use the following command:

```
show configuration
```

The default configuration file is `/opt/vyatta/etc/config/config.boot`

To enter configuration mode, use the following command:

```
configure
```

Steps to configure Vyatta as per Figure 1: Let `eth0` be the internal interface and `eth1` be the WAN facing interface. We will configure interface `eth0` as `192.168.51.1/24` and use `121.x.x.2/30` for `eth1`.

```
set interfaces ethernet eth0 address 192.168.51.1/24
set interfaces ethernet eth1 address 121.x.x.2/30
```

Please note that you can specify the subnet mask in CIDR notation.

Use the `show interfaces` command and verify the assigned addresses by the command. Please also try entering `show i` and observe the result.

Continue the configuration by setting the system default